B1 passage fusion with LoRMA+: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

A judgment is longer than one encoder context, so use passages and fuse their evidence before generation. Document-only uniform coverage including the first and last passage avoids choosing test passages from reference summaries.

Architecture and mechanism

The Phase 1 B1 encoder represents passages; a fusion module composes them for an autoregressive decoder. LoRMA+ adapts output-side transformations while the inherited weights remain the starting point.

Measured result

Scored submission	1030742, 1030744, 1030983
Codabench BERTScore F1	0.7674
Archived local metric	-1.6703 validation negative NLL; BERTScore not measured in this report

The scored BERTScore F1 is 0.7674. The saved report selected epoch 6 by negative summary NLL, not by test or Codabench score.

Limitations and next check

Semantic similarity does not by itself establish factual or legal correctness of a generated summary. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: b1_fusion.py, phase6_data.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.